# NOVELTY & PRIOR-ART ASSESSMENT

Directional averages of 3D contour generators; Willmore-type bounds; genus-dependent extremal problems

**Review date:** 6 November 2026 | **Scope:** mathematical novelty, not authorship or publication priority certification

**Executive assessment:** The supplied manuscript contains mathematically defensible statements, but its original Willmore inequality is an elementary consequence of a coarea identity, pointwise ellipse estimates, Cauchy–Schwarz, and Gauss–Bonnet. There is extensive prior work on averaged silhouettes, Cauchy/Minkowski projection formulas, and curvature integrals. We did **not** find the exact unprojected formula stated verbatim in the targeted searches; absence from these searches does not imply novelty. The best opportunity is a genuinely nontrivial, sharp **fixed-genus optimization or stability result**.

**Recommendation:** Do not submit the existing six-page revision as a stand-alone novel research theorem yet. Treat it as a correct preliminary lemma or expository note while developing a stronger claim with verified priority.

## 1. Mathematical object and the claims under review

For a smooth, compact, boundaryless oriented surface S in three-dimensional Euclidean space, set $\Gamma(u) = \{p \text{ in } S : N(p)\cdot u = 0\}$. The mean **spatial** length of the complete (possibly hidden) contour generators is $L = (1/4\pi)\int_{S^2} \text{length}(\Gamma(u))\, du$. Write A for area, $W = \int_S H^2\, dA$ for the Willmore energy with $H = (\kappa_1+\kappa_2)/2$, and $\chi$ for Euler characteristic.

$$L^2 \le (A/2)(W - \pi\chi)$$

The original claim is sharp globally because spheres attain equality. The exact coarea representation is $L = (1/4\pi)\int_S\int_0^{2\pi} \sqrt{(\kappa_1^2\cos^2\theta + \kappa_2^2\sin^2\theta)}\, d\theta\, dA$. It is a natural instance of the coarea formula for a Gauss-map incidence relation, which significantly reduces the expected novelty of the derivation.

## 2. Closest and most consequential prior art

**[1] R. E. Miles (1975):** An explicit derivation of Minkowski-type mean-projection identities for convex sets. In particular, random line-projection lengths are expressed through total mean curvature. This puts convex projected-contour formulas squarely within classical integral geometry, although they are not literally the same as an unprojected rim-length formula.

**[2] M. McGuire (2004):** Expected silhouette-edge counts of polyhedral models are related to sums of dihedral angles. This provides early and direct evidence that silhouette expectations over directions are established research objects.

**[3] M. Glisse and S. Lazard (2008):** An Upper Bound on the Average Size of Silhouettes, Discrete & Computational Geometry 40, 241–257. The work studies average silhouette lengths in the setting of nonconvex surfaces and establishes expected O(√n) mesh-silhouette complexity under geometric approximation hypotheses. The paper distinguishes transparent spatial rims and projected silhouettes, and uses average silhouette length explicitly. It is strong contextual prior art, **not an identified publication of the exact inequality at issue**.

**[4] S. Plantinga and G. Vegter (2003):** Established smooth contour-generator terminology, projected apparent contours, generic singularities, and robust contour algorithms.

**[5] I. Izmestiev (2022):** A published survey of mixed volumes and convex inequalities, including the proportionality of total mean curvature and mean width and classical Cauchy/Crofton identities. The convex case requires a careful comparison to this standard theory.

## 3. Claim-level novelty ledger

| Claim | Assessment / priority risk |
|---|---|
| **Exact spatial coarea formula** | Not found verbatim in targeted searches. Very natural general coarea identity, so limited intrinsic novelty; priority remains unverified. |
| **Sharp Willmore bound and sphere equality** | Correct under the manuscript assumptions, but follows immediately from standard estimates. Low mathematical depth and likely weak as a stand-alone paper. |
| **Projected length formula, convex case** | Essentially classical projection/total-mean-curvature geometry (Miles/Minkowski/Cauchy). Not an appropriate independent novelty claim. |
| **Projected length with multiplicity, nonconvex case** | Potentially useful explicit representation. It is a straightforward weighted coarea calculation; exact bibliographic antecedent unresolved. |
| **Fixed-genus optimal coefficient c(g)** | Plausible research direction with nontrivial variational content. Its open status has NOT been established; review further before claiming a new problem. |

## 4. A critical false proposal and a correct sharp replacement

The proposed genus-independent inequality $L^2 \le AW/4$ is **false**; likewise the supporting suggestion $\int|H|\,dA \ge 2L$ is false even among convex bodies. A long smoothly rounded cylinder provides explicit asymptotics.

For a cylinder of radius r and lateral length h (smoothly rounded ends of fixed geometry), as $h \to \infty$:

$$\mathbf{L \sim 2h\,; \quad A \sim 2\pi rh\,; \quad W \sim \pi h/(2r).}$$

Therefore $4L^2/(AW) \to 16/\pi^2 \approx 1.62114 > 1$. The candidate $c(0) = 1$ is disproved even for convex genus-zero surfaces.

**Provable convex replacement:** for nonnegative principal curvatures, the curvature-ellipse perimeter obeys $P(\kappa_1,\kappa_2) \le 4(\kappa_1 + \kappa_2)$, since $\sqrt{(a^2\cos^2\theta + b^2\sin^2\theta)} \le a|\cos\theta| + b|\sin\theta|$. Consequently $L \le (2/\pi)\int H\,dA$ and Cauchy–Schwarz yields

$$\mathbf{L^2 \le (4/\pi^2)\,AW \text{ (convex } S).}$$

The long rounded-cylinder family makes the constant $4/\pi^2$ sharp within the convex class. This is a useful correction, but its proof uses elementary estimates; a separate prior-art check is needed before calling the exact coefficient new.

## 5. A better-formulated extremal programme

For each nonnegative integer g, define the dimensionless extremal constant over connected smooth embedded closed surfaces of genus g:

$$\mathbf{c(g) := \sup\{\, 4L(S)^2 / [A(S)W(S)] : \text{genus}(S)=g \,\}.}$$

The sphere gives ratio 1. The rounded-cylinder family gives **$c(0) \ge 16/\pi^2 \approx 1.62114$**. A ring torus with major/minor radii (R,r)=(2,1), based on the manuscript's numerically integrated contour length $L \approx 28.233659$ and exact $A=8\pi^2$ and $W=4\pi^2/\sqrt{3}$, gives **$c(1) \ge 1.77176$**. These are lower bounds, not exact values.

The original inequality gives $4L^2/(AW) \le 2[1 + 2\pi(g-1)/W]$. Thus $c(0), c(1) \le 2$; and for $g \ge 1$, $W \ge 4\pi$ gives $c(g) \le g+1$. None is known here to be optimal.

**Meaningful potential advances:** prove an improved upper bound for c(0) or c(1); identify an extremizing or maximizing-sequence geometry; prove quantitative stability near the sphere in a suitably energy-controlled class; or obtain genuinely visibility-aware (occlusion-sensitive) contour estimates. A numerical optimization result alone is a conjecture generator, not a proof.

## 6. Publication assessment and safeguards

| Dimension | Referee assessment |
|---|---|
| **Correctness** | Central inequality and the corrected coarea proof: sound within the stated smooth compact setting. |
| **Originality** | Low to uncertain for the existing theorem. Specific identity not ruled out as previously known. |
| **Potential strength** | Moderate if a genuinely new sharp c(g) theorem, stability theorem, or nonlocal visibility bound is proved. |
| **Readiness** | NOT ready as a claimed novel stand-alone research article. |

**Necessary checks before submission:** (i) search MathSciNet/zbMATH and citation networks using “transparent silhouette”, “rim”, “profile curve”, “Gauss map inverse equators”, “mean shadow perimeter” and “contour generator length”; (ii) compare precise object definitions and normalization constants; (iii) disprove candidate inequalities on cylinders, tori and high-curvature necks before numerical optimization; (iv) include the Glisse–Lazard, Miles, McGuire and convex-geometry references; and (v) resolve human authorship and originality attribution for the AI-audit additions.

## 7. References / verified bibliographic leads


1. R. E. Miles, “Direct derivations of certain surface integral formulae for the mean projections of a convex set,” Advances in Applied Probability 7 (1975), 818–829. DOI: 10.2307/1426401. https://doi.org/10.2307/1426401

2. M. McGuire, “Observations on Silhouette Sizes,” Journal of Graphics Tools 9(1) (2004), 1–12. DOI: 10.1080/10867651.2004.10487594. https://doi.org/10.1080/10867651.2004.10487594

3. M. Glisse and S. Lazard, “An Upper Bound on the Average Size of Silhouettes,” Discrete & Computational Geometry 40 (2008), 241–257. DOI: 10.1007/s00454-008-9089-3. https://arxiv.org/abs/cs/0702087

4. S. Plantinga and G. Vegter, “Contour Generators of Evolving Implicit Surfaces,” University of Groningen proceedings, 2003. https://research.rug.nl/en/publications/contour-generators-of-evolving-implicit-surfaces/

5. I. Izmestiev, “Convex Bodies: Mixed Volumes and Inequalities,” in Surveys in Geometry I (2022), 171–203. DOI: 10.1007/978-3-030-86695-2_5. https://doi.org/10.1007/978-3-030-86695-2_5

6. E. W. Weisstein, “Cauchy’s Perimeter Formula,” MathWorld. https://mathworld.wolfram.com/CauchysPerimeterFormula.html

7. The audited manuscript: I. Bədəlov, “Directional Averages of Contour Generators on Closed Surfaces: A Sharp Willmore-Type Bound and a Projected-Length Identity,” AI-assisted audit revision, 4 November 2026 (user-supplied, unpublished).


**Limits of review:** This is a targeted public-web prior-art review supplemented by mathematical checking. It is not an exhaustive MathSciNet/zbMATH database examination, does not certify first publication, and cannot serve as a legal patent novelty opinion or an independent peer-review decision.